



University of Essex
Department of Mathematical Sciences

MA981: DISSERTATION

Improving Classification model for chronic kidney disease dataset by Feature selection

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Introduction

1.1 Research Background

Chronic Kidney Disease (CKD) is a significant global health issue affecting patients with non-communicable diseases. It is one of the leading non-communicable causes of death worldwide. The prevalence of CKD continually increases worldwide. Chronic kidney disease (CKD) can occur more frequently in risk groups, such as specific age brackets, genders, or races. [1]

The definition of CKD is decreased kidney function shown by glomerular filtration rate (GFR) of less than 60 mL/min, or markers of kidney damage, or both, of at least three months duration, regardless of the underlying cause. Diabetes and hypertension are the main causes of CKD in all high-income and middle-income countries and many low-income countries.[2] Furthermore, CKD can progress to kidney failure, necessitating dialysis or a kidney transplant. It is easier to treat and manage CKD in its early stages. Early detection of CKD is crucial for physicians to provide timely and effective treatment. For this reason, it is important to identify early-stage CKD patients as it can lead to more effective treatment.[3]

Machine learning algorithms can assist in accurately labeling patients to determine who is likely to have CKD. [4] Additionally, improving the model's performance is essential for healthcare providers to make better decisions. Machine learning requires an adequate number of samples to prevent overfitting while avoiding an excessive number of features that could create a more complex algorithm. [5]

Feature selection is one of the significant methods used to reduce noise and eliminate redundant data, enhancing the model's performance. It is the one successful method to solve problems in the healthcare industry. It not only reduces irrelevant features but also reduces cost. [6] Feature selection chooses a subset of relevant features from the original dataset. [7]

For this reason, we will experiment with feature selection to improve the classification model for CKD.

1.2 Research Aim and Scope

This study aims to improve the effectiveness of a classification model for CKD using feature selection methods. We will discuss feature selection to improve the model for predicting CKD diagnosis in the UCI chronic kidney disease dataset. The UCI dataset has 400 observations with 24 attributes and one target. The chosen machine learning classifiers, Support Vector Machine, Naive Bayes, Logistic Regression, and K-Nearest Neighbours will be employed.

1.3 Research Methodology

The classification is implemented using Python 3.3, leveraging several libraries such as Numpy, Pandas, and some scikit-learn. Feature selection is applied to the dataset and subsequently employed with each algorithm. The study compares the performance of the models before and after applying feature selection using each method. Each model, with or without the feature selection technique, is evaluated based on accuracy, precision, recall, and F1 score.

The flowchart of this study can be illustrated as [1.1](#).

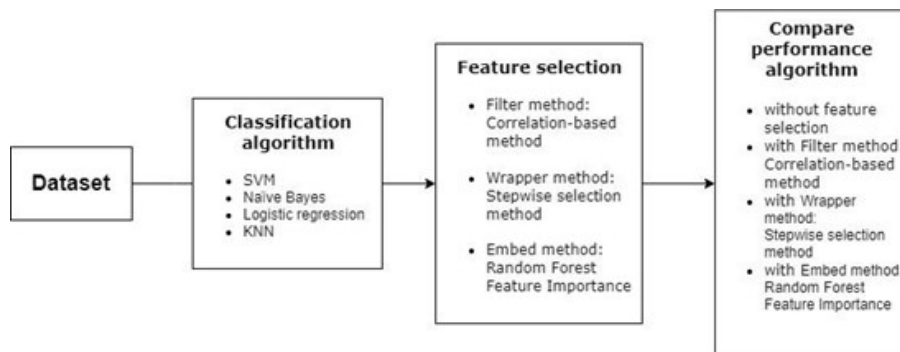


Figure 1.1: The flowchart of this study

1.4 Dissertation outline

This paper is organized as follows: Section 2 presents previous studies related to chronic kidney disease classification and feature selection. Sections 3,4, and 5 cover the knowledge about chronic kidney disease, the feature selection technique, and the classifier, respectively. In Section 6, we provide details about the dataset and the experiment. Section 7 presents and discusses the results. Finally, Section 8 concludes the study with recommendations concerning the dataset and methodology.

Literature review

There are many studies about using various machine learning algorithms to improve the classification of chronic kidney disease.

Debal D and Sitote T conducted a comparison of predictive models, such as Random Forest (RF), Support Vector Machine (SVM), and Decision Tree (DT), with feature selection. They employed these models on a dataset of approximately 1718 CKD inpatients during 2018-2019. The dataset comprises 19 features and one multiclass target, which provides information about the stage of CKD. The study revealed that a random forest based on recursive feature elimination using cross-validation exhibited the best performance. [8]

Zeynu.S improved the CKD classification algorithms such as K-Nearest Neighbor, J48, Artificial Neural Network, Naïve Bayes, and Support Vector Machine by feature selection and ensemble technique. The UCI chronic kidney disease has 400 observations with 24 attributes and 1 binary target, presence of CKD or not. The result is the ensemble improves the classification algorithm better than other methods.[9]

Salekin.A and Stankovic.J compare the feature selection, wrapper method and embedded methods, and LASSO regularization, to improve Random Forest, KNN, and Neural networks. The results show Random Forest with the wrapper method set at 12 features performs best.[10]

Jongbo O and Olowookere T, in collaboration with Adetunmbi, investigated the influence of a bagging ensemble technique on the CKD dataset. They utilized an efficient method for selecting relevant features to enhance the efficiency of predictive models. The examination

was conducted using the CKD dataset from the UCI Machine Learning Repository. The study's findings revealed that the combination of feature selection and ensemble techniques considerably elevated the accuracy of data mining classification algorithms.[11]

Polat.H,Danaei Mehr.H, Cetin.A used 2 feature selections, including the wrapper and filter methods, to improve the Support Vector Machine classifier (SVM). The result indicated the Support Vector Machine classifier, when combined with the filtered subset evaluator and Best First search engine is the highest accuracy at 98.5% [12]

Kotturu.P, Sasank.V, Supriya.G, Manoj.C, and Maheshwarredy.M conducted a comparison of classifiers using the CKD UCI dataset, specifically focusing on decision tree and support vector machine. The findings reveal that the support vector machine outperformed the decision tree, achieving accuracy rates of 96.75% and 91.75%, respectively.[13]

Chronic kidney disease

3.1 Chronic Kidney Disease Prevalence

The prevalence of chronic kidney disease (CKD) is a major factor in death rates across the globe. There were around 843.6 million CKD patients worldwide as of the year 2017. This represents a sizeable 10 percent of the world's population.[14] As a result, by the year 2019, the prevalence of CKD had increased to an average of 13.4 percent, raising concerns about this medical condition. Particularly, the number of patients requiring kidney replacement therapy increased globally, from 4.902 to 7.083 million patients.[15] The major causes of CKD, including diabetes mellitus, hypertension, obesity, and advanced age, all contribute to this worrying trend. Surprisingly, it has been discovered that about 10 percent of adults struggle with the effects of this illness, causing an alarming 1.2 million deaths annually and an amazing 28.0 million years of life to be cut short prematurely. This sobering truth makes CKD a significant contributor to the pattern of world death.[14] In consideration of, CKD emerges as a sentinel of global mortality.

Furthermore, countries with low and middle-income economies are impacted by the incidence of chronic renal disease. It often significantly impacts people prone to higher risks, especially those dealing with issues like diabetes, hypertension, old age, and even gender-based inequities. As the future approaches, it is becoming increasingly essential to give CKD prevention and decrease its incidence priority. If such interventions are not implemented, CKD will likely rise to become the fifth leading cause of mortality by 2040. [1]

Awareness and intervention become more critical in this combination of occurrences. Predicting the progress of CKD is crucial because it has the possibility of preventing the disease's insidious progression and giving individuals stuck in its traps an enhanced standard of life. Furthermore, as an outcome of integrating comprehension, prediction, and intervention in the ongoing struggle against chronic kidney disease, this foresightful endeavor promises to produce substantial financial benefits within the healthcare sector.

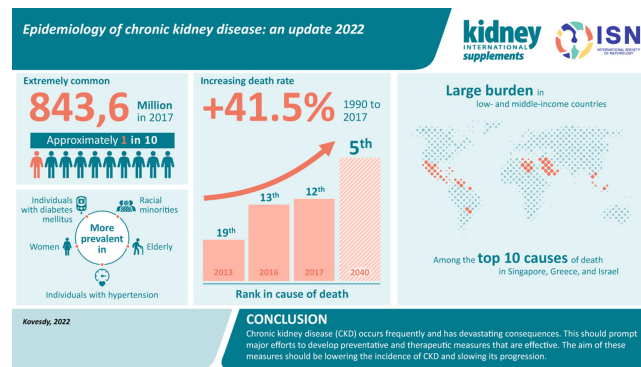


Figure 3.1: The epidemiology of chronic kidney disease 2022

[1]

3.2 Definition of Chronic kidney disease

Chronic kidney disease (CKD) is a progressive disease that reduces kidney function. It is characterised by a long-term, steady reduction in kidney function, typically reflected in a lower glomerular filtration rate (GFR), a crucial measure of the kidneys' capacity to remove waste and extra fluid from the blood. The diagnostic criteria for CKD require the persistence of a low estimated GFR (eGFR) for at least 90 days, which is used to categorise the disease into 5 stages. These stages assist medical practitioners in determining the severity of the illness and developing suitable treatment methods.

Based on the eGFR, a computed value that considers characteristics including age, sex, race, and serum creatinine levels, the stages of CKD are categorised. The staging system directs the treatment of the disease and offers helpful perspectives into the severity of kidney damage. The stages you mentioned are further explained as follows:

Stage 1 CKD: In this stage, the eGFR is more than 90 mL/min per 1.73 m². Although kidney function may be slightly reduced, individuals in this stage often do not exhibit

significant symptoms. However, it is an early indicator of potential kidney issues and should be closely monitored.

Stage 2 CKD: This stage is characterized by an eGFR between 60 and 90 mL/min per 1.73 m^2 . While kidney function is further reduced compared to Stage 1, individuals may still not experience noticeable symptoms. Monitoring and early intervention are essential to slow down the progression of the disease.

Stage 3 CKD: In Stage 3, the eGFR ranges from 30 to 60 mL/min per 1.73 m^2 . Kidney function is noticeably impaired, and symptoms such as fatigue, fluid retention, and changes in urine output may become more apparent. Medical management becomes increasingly important at this stage.

Stage 4 CKD: With an eGFR of 15 to 29 mL/min per 1.73 m^2 , Stage 4 indicates severe kidney dysfunction. Individuals may experience significant symptoms, such as high blood pressure, anemia, bone problems, and electrolyte imbalances. Close medical supervision and potentially more aggressive interventions are required to manage complications.

Stage 5 CKD (End-Stage Renal Disease, ESRD): The most advanced stage is where the eGFR drops below 15 mL/min per 1.73 m^2 . Kidney function is severely compromised, and individuals in this stage require renal replacement therapy, such as dialysis or kidney transplantation, to sustain life.

It's important to note that CKD is a complicated condition affected by various factors, including genetics, lifestyle choices, and underlying health conditions. Early detection, lifestyle modifications, and appropriate medical care can help decrease the progression of CKD and manage its associated complications. Regular monitoring of kidney function through eGFR assessments and collaboration with healthcare providers are crucial for effectively managing this chronic condition. [1]

3.3 Risk factor of Chronic kidney disease

The probability of getting Chronic Kidney Disease (CKD) is impacted by a wide range of factors leading to a complex combination of several causes. These risk indicators can be usefully categorized into various categories, each of which reveals a different aspect of the condition:

1. **Demographic Factors:** Many demographic factors are directly related to CKD risk.

One of these is advanced age when the risk gradually increases as people age. Additionally, race can have a significant impact, with certain ethnic groups being more vulnerable than others. Furthermore, risk factors for an individual's health progression may include low birth weight, which is frequently indicative of early-life health issues, and a family history of CKD.

2. **Lifestyle Factors:** Individual choices about lifestyle have a significant impact on CKD risk. Smoking and being overweight have been related to a higher risk of developing CKD. In addition, heavy alcohol intake increases the risk of developing CKD and its immediate adverse health effects. If not used carefully, the use of analgesic drugs might further increase the risk factor.
3. **Medical issues:** CKD risk can be considerably increased by various medical problems. Uncontrolled hypertension has been recognized as a critical risk factor and is a common health concern. Critical medical variables include uncontrolled diabetes mellitus, cardiovascular disease, hyperlipidemia, and metabolic syndrome. Due to the complicated associations between CKD risk and the hepatitis C virus and HIV infections, their presence adds another difficulty level. Even cancer can increase the vulnerability to CKD due to its systemic effects.
4. **Other Factors:** Besides the previously mentioned factor, exposure to heavy metals might seem to be a significant risk factor. External forces and environmental factors can significantly impact how CKD develops.

These risk variables represent a complicated combination of factors that mutually increase the possibility of developing CKD. Knowing about and managing these factors is essential for risk assessment and for developing successful management and preventative strategies for reducing the effects of CKD.[16]

Feature selection

A high number of features in a dataset leads to model overfitting, leading to poor results on the validation datasets. Additionally, constructing models from datasets with many features is more computationally demanding. The feature set reduction is performed through feature extraction (transformation) and feature selection.[17] The dissimilarity between these two approaches resides in their strategies: feature extraction combines the original dataset to produce new and lower-dimensional features. In contrast, feature selection focuses on identifying a subset from the original dataset that effectively minimizes redundancy and maximizes its pertinence to the target variable. [7]

Feature selection can be categorized into supervised, unsupervised, and semi-supervised algorithms. The supervised method employs the relevant features from labeled data. It can be divided into filter, wrapper, and embedded models. Filter models evaluate features without using any classification algorithm. They rank features based on specific criteria such as coefficient correlation, Fisher score, or Relief. Subsequently, the features with the highest ranking will be chosen to be applied to the classification model. The wrapper model utilizes the predictive accuracy of a predetermined learning algorithm to determine the quality of selected features. However, the wrapper method is likely more accurate than the filter method. However, they have to run the classifier many times to evaluate the chosen subset features, which comes with a higher computational cost when dealing with numerous features. The embedded model combines aspects of both the filter and wrapper methods. It employs statistical criteria to choose features similar to the filter model and selects the subset

with the highest classification accuracy, akin to the wrapper method.[7]

The unsupervised feature selection method is a less constrained search problem without class labels, depending on cluster quality measures, and can eventuate many equally valid features subset, but It is hard to recover removed features without additional conditions. Semi-supervised feature selection uses labeled and unlabelled data to estimate feature relevance. [7]

Medical data consists of a large number of dimensions, and to reduce this complexity, feature selection is employed. This approach involves selecting specific features based on relevance evaluation criteria, resulting in improved learning performance, reduced computational expenses, and enhanced model interpretability.[7]

4.1 Filter method: Correlation Coefficient based method

The filter methods select features based on a performance measure regardless of the employed data modeling algorithm. A typical filter algorithm consists of two steps. The first step ranks individual features or evaluates features subset based on certain criteria. Then, the highest-ranking features will be chosen for classification models. [7] The filter method measurements are information, distance, consistency, similarity, and statistical measures. The filter methods are classified based on classification, regression, and clustering tasks. For instance, Correlation-based Feature Selection (CFS) is well-suited for multivariate datasets and employs statistical techniques. [17]

4.2 Wrapper method: Forward and Backward selection

The wrapper method considers feature subsets by the performance quality of a modeling algorithm.[17] Wrapper models obtain better predictive accuracy estimates than filter models, but they are much more computational cost. Regarding the wrapper model, the final set of features will be from searching algorithms such as hill-climbing, best-first, branch-and-bound, and genetic algorithms. They come in two flavors, forward selection and backward selection.[7]

The first two searching techniques in the wrapper methods described in the subsequent section are the two most common greedy methods frequently employed for feature

selection.[18]

Forward selection: This method start from the empty set. It performs best when only a small number of features are involved. However, the drawback of this method is it can not remove features that become insignificant after adding them before.[18]

Backward selection: This method begins with the complete set of features, then elimination works best with many features in the dataset.[18]

Forward and Backward selection: This method allows dynamic addition and deletion of the features subset until the best subset is obtained. It is probably the most compelling feature selection method.[18]

4.3 Embedded method: Rain Forest method

Within feature selection methods, embedded models represent a major approach. Embedded models smoothly incorporate feature selection into the model training process, unlike independent techniques that perform independent feature assessment or employ various models for evaluation. This integration produces more efficient and effective results by enabling models to simultaneously learn from the data and highlight the most important traits.[7]

Embedded techniques combine feature selection with model training. Models improve their predicting ability as they improve their understanding by adjusting feature importance based on the distinctive features of the data. Informing one another, picking characteristics, and training a model simultaneously can increase efficiency. It also adjusts its feature selection technique in response to the complexities of the classification task, enabling the model to adapt to different levels of complexity.[7]

Classification model

Classification is a process that predicts qualitative responses for observation by assigning them to a specific category or class. The classification model often utilizes probabilities to predict observations within classes. In this study, we employ four classification techniques, logistic regression, support vector machine, Naive Bays classifier, and K-nearest Neighbours classifier, to compare with the dataset, we apply feature selection. [19]

5.1 Logistic Regression

Logistic regression applies probability principles to build a relationship between input features and the response variable, Y . Its primary use is in binary classification problems, where classes are indicated by the numbers "1, 0." The basic idea is to calculate the probability $p(X) = p(Y = 1 | X)$ using input features (X). This estimation is based on the presumption that each answer, $Y_1, Y_2, \dots, Y_n$, follows an independent distribution, $\text{beta}(1, p(X_i))$, where $x = (x_1, \dots, x_n) \in R_n$.

The logistic regression model is defined by

$$p(x_i) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \quad (5.1.1)$$

The maximum likelihood technique is used to fit this model. Equation 5.1.1 demonstrates that the resulting function regularly produces an S-shaped curve, which is a distinguishing feature. It is noticeable that the projected probabilities are always in the interval between 0

and 1, characteristic of its behavior.

In conclusion, logistic regression provides an effective approach for analyzing classification probabilities. This methodology creates a predictive framework corresponding to the S-shaped curve by connecting attributes to responses through probabilities, ensuring that predicted probabilities remain within a useful range.[19]

5.2 Support Vector Machine

The Support Vector Machine (SVM) is an efficient classifier that successfully separates observations using hyperplanes. In binary classification tasks, this method is recognized for its accuracy. SVM achieves a linear boundary by using a separating hyperplane.

The maximal margin hyperplane is crucial in its fundamental form. This hyperplane has been strategically placed to be the furthest away from training observations to maximize separation. It determines the shortest distance, margin, or measure of proximity between observations and the hyperplane. Its primary advantage is the classifier's potential to choose the maximal margin hyperplane, which offers noticeable separation.

Although the maximal margin classifier performs well, it should be noted that high-dimensional spaces, where there are a lot of p -dimensional features, can make it prone to overfitting.

The Support Vector Machine is an effective classification method that effectively divides data into sections using hyperplanes. It is distinguished by its pursuit of the maximal margin hyperplane's maximum separation, however, it does so while exercising caution to avoid overfitting in high-dimensional circumstances.

The support vector machine model is defined by[19]

$$\text{maximize}_{\beta_0, \beta_1, \dots, \beta_p, \epsilon_1, \dots, \epsilon_n, M} \quad M \quad (5.2.1)$$

$$\text{subject to} \quad \sum_{j=1}^p \beta_j^2 = 1, \quad (5.2.2)$$

$$y_i(\beta_0 + \beta_1 x_{i1}) + \beta_2 x_{i2} + \dots + \beta_p x_{ip} \geq M(1 - \epsilon_i) \quad (5.2.3)$$

$$\epsilon_i \geq 0, \sum_{i=1}^n \epsilon_i \leq C, \quad (5.2.4)$$

5.3 Naive Bayes classifier

The naive Bayes classifier, which uses the Bayes theorem to calculate the posterior probability of a class, is an effective classification tool. the Bayes theorem can state by

$$P(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)} \quad (5.3.1)$$

Assigning instances to the class with the highest probability is what this classifier does after calculating probabilities for each class. It can reduce while maintaining efficiency by assuming that characteristics are independent given the class label. The naive Bayes classifier's combination of simplicity and effectiveness gives it its refined appearance. Machine learning deals with complex classification challenges while adhering to the principle of feature independence by applying probability theory. This persistent strategy is fundamental, connecting data and mathematics to produce accurate classification across various purposes.

The Naive Bayes classifier is defined by

$$p(Y|X) = p(X|Y).p(Y)p(X) \quad (5.3.2)$$

The naive Bayes classifier essentially demonstrates the effectiveness of probabilistic reasoning. With the Bayes theorem as its basis and assuming feature independence, it shows how fundamental ideas and actual data may work together to produce impressive classification results. [19]

5.4 K-nearest Neighbours classifier

K-nearest Neighbours (KNN) classifier is the method that estimates the conditional distribution of Y given X and then classifies a given observation to the class with the highest *estimated* probability. Initial with a positive integer K and a test observation x_0 , the KNN classifier identifies the K points in the training data closest to x_0 , represented by N_0 . Then, the

conditional probability for class j is estimated as the fraction of points in N_0 whose response value equal j :

$$P(Y = j|X = x_0) = \frac{1}{K} \sum_{i \in N_0} I(y_i = j) \quad (5.4.1)$$

Eventually, KNN classifies the test observation x_0 to the class with the most significant probability from 5.4.1 [19]

5.5 Metrics for classification model

A performance metric for a classification model that involves an unbalanced dataset. Examine the issues raised by imbalanced datasets, when one class dominates the other by a large margin and causes affected learning patterns in machine learning models. The three types of metrics that can be grouped together are threshold, ranking, and probabilistic metrics. Although probabilistic measures occasionally perform well in imbalanced domains, this performance pattern is not caused by their probabilistic nature.[20]

The number of samples from the majority class divided by the number of samples from the minority class is known as the imbalance ratio (IR), and it can vary from less than 5 (slightly imbalanced), between 9 and 13 (moderately imbalanced), and greater than 16.9 (heavily imbalanced). [20]

The performance metrics for binary classification models are crucial for the accurate assessment of binary classification models. Choosing an appropriate metric should align with the specific context and objectives of the task. [21]

Accuracy: The ratio of correctly predicted instances to the total number. Its reliability can be compromised in datasets with imbalanced class distributions.[21]

Precision: It is referred to as positive predictive value. Precision focuses on the accuracy of positive predictions. It reveals the proportion of predicted positive instances belonging to the positive class.[21]

Recall (Sensitivity): This metric evaluates a model's capability to identify all relevant instances within the positive class. It highlights the ratio of correctly predicted actual positive instances.[21]

F1-Score: A mean of precision and recall, providing a balanced metric that factors in false positives and false negatives.[21]

Selecting the most fitting metric based on the specific context and goals is crucial. By understanding the subtleties of these metrics, practitioners and researchers can accurately evaluate model performance and make informed decisions in real-world applications.[21]

Data visualization and Methodology

6.1 Data

This dataset is the Chronic Kidney Disease (CKD) open dataset from the UCI Machine Learning Repository, 2015. It is a multivariate dataset containing 400 observations and 25 mixed-type features, including 11 numeric features and 14 nominal features such as albumin, DM, hypertension, etc. The target variable of this dataset, named "classification," is binary, representing the presence or absence of CKD. It consists of 250 instances with CKD presence, accounting for 62.5 percent, and 150 instances without CKD, constituting 37.5 percent. [22] More information about these attributes is presented in the table 6.1.

6.2 Data Visualization

In this section, we present the representation of the UCI CKD dataset to gain insights and identify correlations between features, contributing to a better understanding of the purpose of this study.

It is crucial to assess the data quality to identify the limitations of this study. As shown in Figure 6.1, the presence of missing values becomes evident. Among the attributes, "rbc", "rc", and "wc" exhibit the three highest percentages of missing data, accounting for approximately 30-40 percent. The target variable, labeled "classification," is complete without missing values. However, addressing these missing values through imputation is essential before

No.	Attribute name	Description	Category	Scale
1	Age	Age of the patient in year	Continuous	Years
2	bp	Blood pressure in mmHg	Continuous	mm/Hg
3	sg	Urine specific gravity	Nominal	1.005-1.025
4	al	Albumin	Nominal	0-5
5	su	Blood sugar	Nominal	0-5
6	rbc	Red blood cells	Nominal	Abnormal,Normal
7	pc	Pus cell	Nominal	Abnormal,Normal
8	pcc	Pus cell cumps	Nominal	Abnormal,Normal
9	ba	Bacteria	Nominal	Not present, Present
10	bgr	Blood glucose random	Continuous	mgs/dl
11	bu	Blood urea	Continuous	mgs/dl
12	sc	Serum Creatinine	Continuous	mgs/dl
13	sod	Sodium	Continuous	mEq/L
14	pot	Potassium	Continuous	mEq/L
15	hemo	Hemoglobin	Continuous	gms
16	pcv	Pack cell volumn	Continuous	Pcv
17	wc	White Blood Cell count	Continuous	cells/cumm
18	rc	Red Blood Cell count	Continuous	millions/cumm
19	htn	Hypertension	Nominal	No,Yes
20	dm	Diabetes mellitus	Nominal	No,Yes
21	cad	Coronary artery disease	Nominal	No,Yes
22	appet	Appetite	Nominal	Good,Poor
23	pe	Pedal edema	Nominal	No,Yes
24	ane	Anemia	Nominal	No,Yes
25	classification	Presence of CKD	Nominal	CKD,Not CKD

Table 6.1: Description of the CKD dataset

implementing this dataset. For a representation of the missing values across the other attributes, please refer to Figure 6.1. The figure graphically illustrates the proportion of missing values for each attribute.

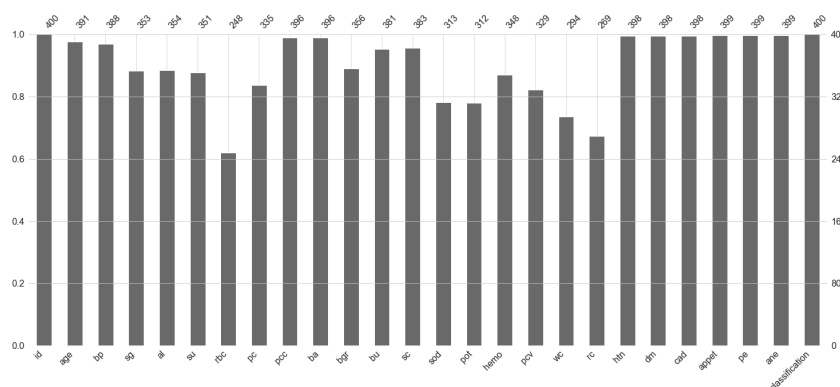


Figure 6.1: Missing value in dataset

Based on the information in figure 6.2, the continuous variables exhibit different scales and are not normally distributed. Some features, such as serum creatinine, potassium, and sodium, have outliers, which are further explained in the figure 6.3, 6.4, 6.5. The distribution showed improvement after imputing the outliers with the mean calculated from data excluding outliers in the figure 6.6

Moving on to categorical variables, we visualized the data using bar graphs in 6.7 and found that the proportions of each variable are not balanced. Additionally, we identified some abnormal values that will need to be addressed.

Based on a review of relevant studies, I have identified key risk factors for Chronic Kidney Disease, including diabetes mellitus, hypertension, advanced age, sex, and ethnicity. However, this dataset only contains information on diabetes mellitus, hypertension, and age. To further explore these relationships, I will create plots to visualize their associations with the "classification".

First, we illustrate the age factor in Figure 6.8 across different age groups. We observed that the proportion of CKD patients is lower in the age group below 50. However, in the age group above 50, the CKD proportion increases by approximately half of the patients.

Regarding hypertension patients, we observed that all individuals with hypertension and diabetes mellitus also have chronic kidney disease (CKD). In contrast, only 50 percent of normal patients are affected by CKD. These findings are presented in Figures 6.10 and 6.9.

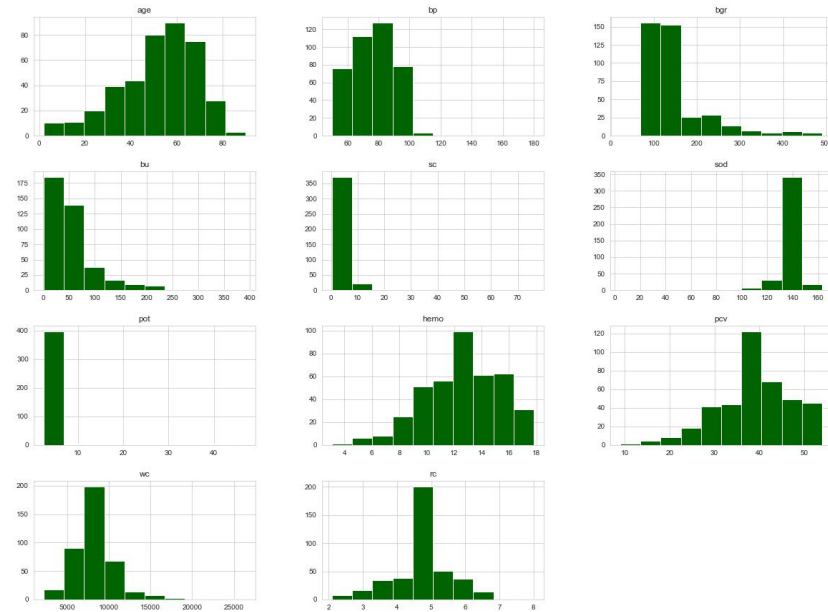


Figure 6.2: The distribution of numerical attributes

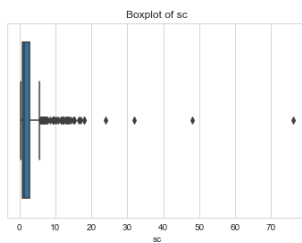


Figure 6.3: The box plot of Serum creatinine (Sc)

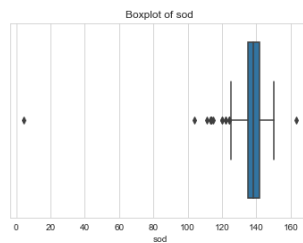


Figure 6.4: The box plot of Sodium (Sod)

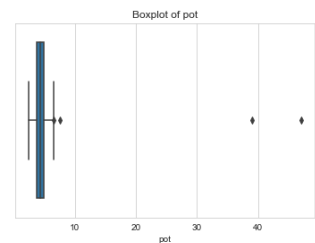


Figure 6.5: The box plot of Potassium (Pot)

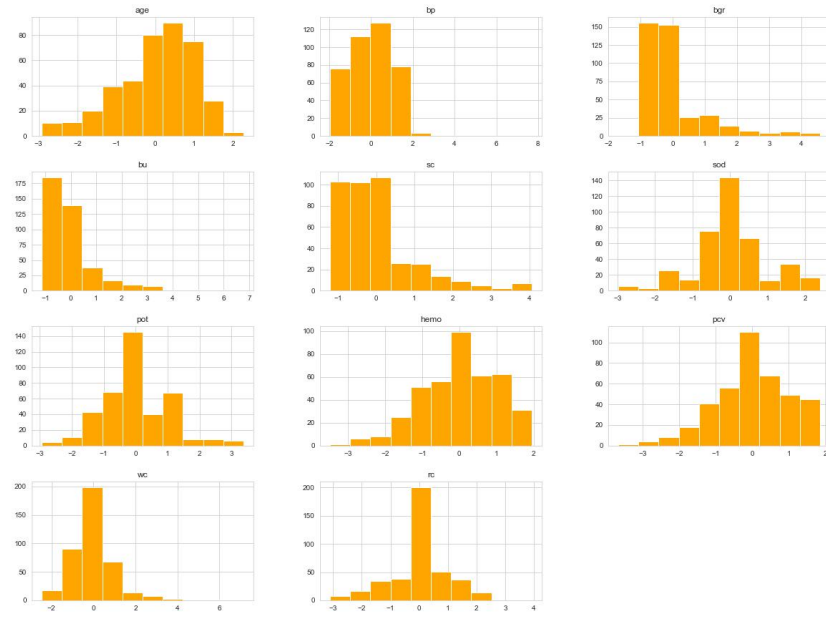


Figure 6.6: The distribution of numerical attributes after scaling data

Our exploratory data analysis identified a correlation between CKD occurrence and risk factors such as advanced age, hypertension, and diabetes mellitus. Patients with these risk factors are more likely to be associated with CKD compared to those without them.

6.3 Methodology

The procedure of this study including 3 stages : (i) data preprocessing (ii) feature selection (iii) classifier application.

6.3.1 Data pre-processing

Data preprocessing is crucial for enhancing data quality and improving the performance of machine learning models. This dataset has missing values, abnormal values, and outliers. In this section, we will explain the pre-processing step for this study.

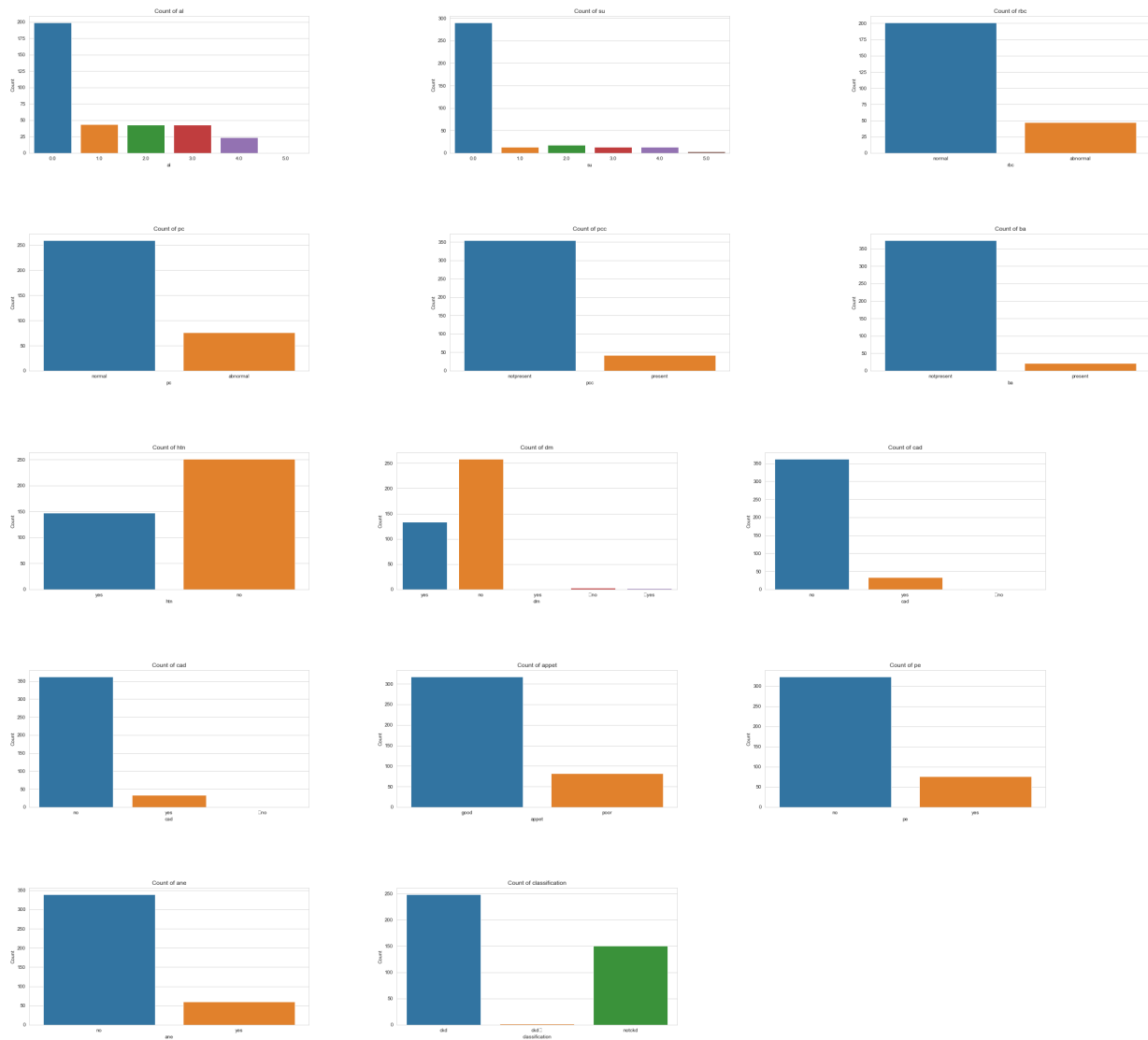


Figure 6.7: The bar graphs of categorical features and target

Abnormal value

We have identified noisy data within this dataset, notably characterized by including unconventional or irregular characters within certain features. A two-pronged approach has been undertaken to rectify this issue and ensure the data's reliability. Firstly, we have initiated a normalization process that encompasses the aforementioned specific features and extends to the entire dataset. This normalization seeks to bring the values of the variables into a standardized range, facilitating more meaningful comparisons and analyses. Secondly, attention has been directed towards three features showing an 'object' data type. Enhancing compatibility and enabling accurate computations, converting these features to the 'float' data type is imperative and suitable for handling numerical values. Furthermore, our observations

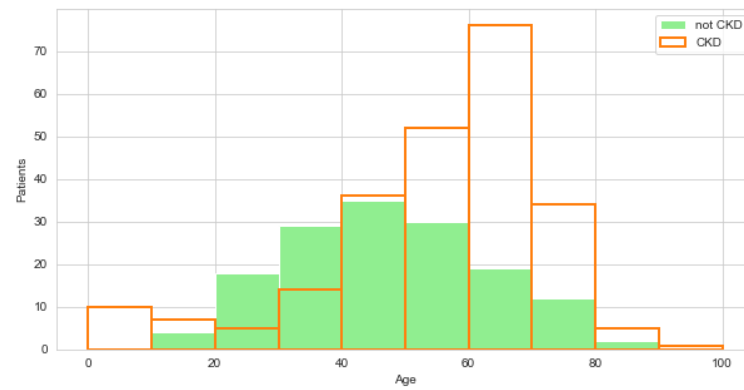


Figure 6.8: The proportion of CKD by aged group

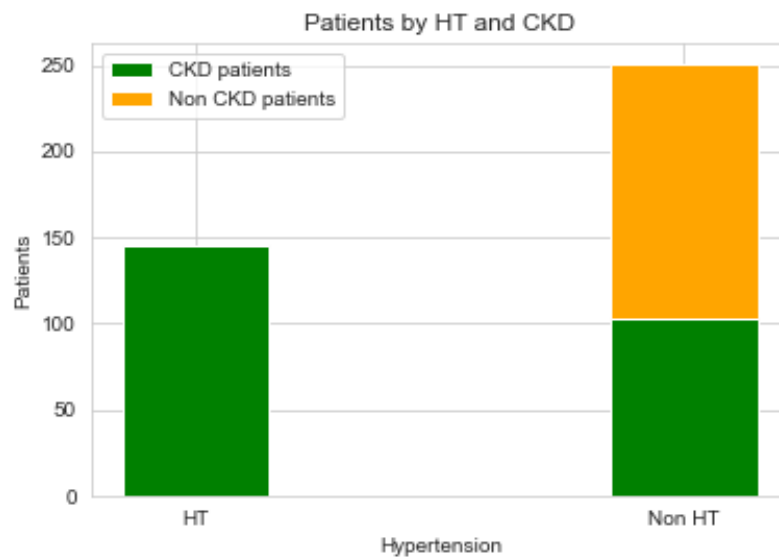


Figure 6.9: The proportion of CKD by hypertension

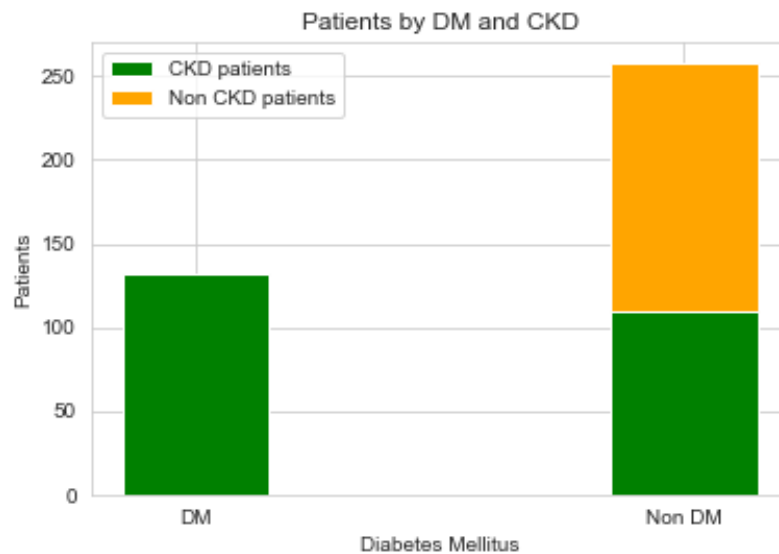


Figure 6.10: The proportion of CKD by diabetes mellitus

from data visualization have unveiled instances of unusual characters and certain attributes. These abnormal characters are necessary for the removal to ensure data integrity and avert any potential distortions, thereby paving the way for a cleaner and more robust dataset for subsequent analysis.

Outlier

We observed that the distribution of serum creatinine (sc), sodium (sod), and potassium (potassium) values within the dataset is less than optimal, indicating potential irregularities or skewness. Delving deeper into these variables, we uncovered the presence of outliers, as visually illustrated in Figures 6.3, 6.4, and 6.5.

To address the issue of outliers and promote a more reliable analysis, we have chosen to employ an imputation strategy. Specifically, we have opted to impute the outliers with the mean value derived from the data points that fall within a defined range. This range is determined by calculating the first quartile (Q1) minus 1.5 times the interquartile range (IQR) as the lower limit, and the third quartile (Q3) plus 1.5 times the IQR as the upper limit. The interquartile range represents the spread of the middle 50

By implementing this approach, we aim to mitigate the influence of outliers on the overall dataset while still retaining the general distribution trends and characteristics. This contributes to a more accurate and informative analysis of the variables in question.

Missing value

Upon reviewing the information presented in Figure 6.1, it becomes evident that notable instances of missing data plague specific attributes in the dataset. Addressing these gaps is crucial for maintaining the integrity and reliability of subsequent analyses. To handle the missing values effectively, I have adopted distinct strategies based on the nature of the features involved. For categorical attributes, where the data consists of discrete categories, the mode is utilized for imputation. This approach is particularly suitable for preserving the categorical nature of the feature and ensuring that the imputed values align with the overall distribution. Conversely, the mean is employed as the imputation strategy for numerical attributes characterized by continuous values. This choice is guided by the desire to maintain the central tendency of the distribution while filling in the missing values. By applying these imputation techniques tailored to the respective feature types, I aim to minimize any potential biases that missing data might introduce, thereby fostering a more accurate and comprehensive dataset analysis.

Scaling

I noticed that the continuous variables within the dataset exhibited variations in their scales, meaning that they were measured in different units or ranges. To facilitate more meaningful comparisons and analyses, I took the step of standardizing these variables. Standardization involves transforming the data in such a way that each variable has a mean of 0 and a standard deviation of 1. This not only brings the variables onto a standard scale but also helps reduce the potential bias introduced by differences in their original magnitudes. By standardizing the scale of these continuous variables, I aimed to ensure that their contributions to the analysis or modeling process are based on their relative relationships rather than on their absolute values.

Encoding

Before implementing the classifier, it is necessary to address the nominal variables within the dataset that require suitable encoding. Nominal variables, often representing categorical data, must be transformed into a numerical format that the classifier can effectively process.

A distinct encoding strategy has been employed in the specific case of categorical features.

Within these features, the number '0' is denoted by utilizing negative indicators such as 'no,' 'abnormal,' and 'not present.' Conversely, when the number '1' is encountered, it is represented directly.

This approach serves a twofold purpose: it facilitates the classification process by converting categorical data into a format amenable to numerical analysis and imparts semantic meaning to the encoding. The encoding becomes more intuitive and interpretable by utilizing explicit negative indicators for the absence of a feature and a straightforward '1' for its presence.

By executing this encoding procedure, we are poised to effectively integrate these nominal variables into the classifier, thus enabling it to make informed and accurate predictions based on the transformed data.

6.3.2 Feature selection

Filter method: Correlation Coefficient base method

In this study, we use coefficient correlation for feature consideration.[7] The process includes the following steps:

1. Calculates feature correlations with a target variable ('classification') based on their correlation values.
2. Create an empty list to store the names of features that strongly correlate with the target variable(above 0.5 or below -0.5).
3. Sort correlation values and their corresponding feature indices: If the correlation is significant, add the feature's name to the list.
4. Select Features: The list contains the names of features that exhibit strong correlations with the target variable.
5. The list contains the names of features that exhibit strong correlations with the target variable.

In summary, this code calculates the correlation between each feature and the target variable, selecting features with substantial correlations (above 0.5 or below -0.5). The chosen

features are stored in the list, facilitating the identification of features strongly connected to the target variable for further analysis or model development.

Wrapper method: Forward and Backward selection

This method performs feature selection with a best-first search using the Sequential Feature Selector (SFS) along with a RandomForestClassifier from scikit-learn. The process involves the following steps:

1. **Create Classifier:** Initialize a RandomForestClassifier instance, which will serve as the base model for evaluation.
2. **Initialize Feature Selector:** Create a SequentialFeatureSelector with parameters specifying to select the best feature subset using forward and backward selection. Use accuracy as the evaluation metric and perform 5-fold cross-validation.
3. **Fit Selector to Data:** Fit the SequentialFeatureSelector on the input data (X) and target labels (y). It identifies the optimal subset of features.
4. **Retrieve Results:** Get the indices of the selected features and extract their corresponding column names along with the target variable from the original DataFrame.

In summary, this method employs SFS to iteratively choose the most important features from the data using a RandomForestClassifier to enhance classification model performance.

Embedded method: Random forest

This method chooses the feature importance scores from the RandomForestClassifier to identify the key features that significantly impact classification accuracy. The process follows these steps:

1. **Create Classifier:** Initialize a RandomForestClassifier to get feature importance scores
2. **Iterative Evaluation of Feature Subsets:** Train the model using your feature matrix X and target vector y to get the feature importance scores attribute of the trained model.
3. **Selecting the Optimal Feature Subset:** Sort the features by importance score and select the top features, then train the model with only selected data.

4. Determine the optimal number of features: Identify the number of features (i) that retrieves the highest accuracy value, then indices of the top features corresponding to the optimal number. Get the optimal subset features with the highest accuracy.

In summary, This method evaluates different feature subsets and measuring accuracy determines the best number of features to use. This strategic approach elevates the model's performance, focusing on the most informative features and enhancing its ability to classify data accurately.

6.3.3 Classifier application

After applying each feature selection method, the classifiers were employed on both the original datasets and the datasets modified with the feature selection methods: filter method, wrapper method, and embedded method. Subsequently, the performance of the models for each dataset was evaluated, facilitating a comparison of the effectiveness of feature selection in enhancing model performance.

Result and Discussion

the results of our feature selection process encompass the count of retained features and the time consumed by each method. Additionally, we provide the outcomes of the classification model derived from diverse feature selection methods. The assessment of model enhancement is gauged through metrics like precision, recall, F1-score, and accuracy.

7.1 The outcome of feature selection

The outcomes of the feature selection process for each method are summarized in table 7.1.

The Filter Method took the quickest only 0.0035 seconds while retaining 8 features. This method works effectively in situations when computational effectiveness is essential.

The Wrapper Method kept the most features at 14, demonstrating its focus on maximizing model accuracy. This required substantial computational time at 643.82401 seconds, however.

The embedded Method maintained 11 features over 78.92151 seconds, balancing computing time and feature retention. It balanced between the Filter and Wrapper methods.

Feature selection	Remained feature	Time(sec)
Filter method	8	0.0035
Wrapper method	14	643.82401
embedded method	11	78.92151

Table 7.1: The outcome of feature selection method

7.2 The performance of model

Moving on to model performance along with the different feature selections, the result is present in table 7.2, 7.3, 7.4, and 7.5.

From the table 7.2, the accuracy of the logistic regression classifier with the original dataset is 0.97 over 0.00279 seconds. After using the wrapper and embedded methods, the accuracy increased to 0.98. In contrast, the filter method's accuracy remained unchanged. The embedded method dataset classifier is the fastest in time, with a 0.00218 seconds response time, while the filter method is the slowest at 0.00364 seconds.

Feature selection (FS) method	Precision	Recall	F1-score	Accuracy	Time(sec)
Without FS	0.97	0.97	0.97	0.97	0.00279
Filter method	0.97	0.97	0.97	0.97	0.00364
Wrapper method	0.98	0.98	0.98	0.98	0.00282
embedded method	0.98	0.98	0.98	0.98	0.00218

Table 7.2: Performance of Logistic regression according to feature selection

From the table 7.3, with the original dataset, the accuracy is 0.97 at 0.00384 seconds. Only the wrapper approach can raise this classifier's accuracy to 0.98; the other's accuracy stays at 0.97. However, the duration spent in the wrapper is the longest by a tiny margin (0.00388 seconds), while the embedded method can decrease the time spent to 0.00237 seconds.

Feature selection (FS) method	Precision	Recall	F1-score	Accuracy	Time(sec)
Without FS	0.97	0.97	0.97	0.97	0.00384
Filter method	0.97	0.97	0.97	0.97	0.00336
Wrapper method	0.98	0.98	0.98	0.98	0.00388
embedded method	0.97	0.97	0.97	0.97	0.00237

Table 7.3: Performance of Support vector machine according to feature selection

From the table 7.4, all Feature Selection techniques influenced the model's performance. All FS approaches showed a slight decrease in accuracy from 0.96 in the original dataset to 0.93 while maintaining consistency in precision, recall, and F1-score. The use of Feature Selection techniques significantly decreased computation time. Comparing the time consumption of these methods to the baseline, the Filter, Wrapper, and embedded methods were efficient.

Feature selection (FS) method	Precision	Recall	F1-score	Accuracy	Time(sec)
Without FS	0.96	0.96	0.96	0.96	0.00531
Filter method	0.95	0.93	0.94	0.93	0.00271
Wrapper method	0.94	0.93	0.93	0.93	0.00274
embedded method	0.95	0.93	0.94	0.93	0.00328

Table 7.4: Performance of Naive Bayes classifier according to feature selection

All feature selection techniques from the table 7.5 improve the model's performance. Precision, recall, and F1-score have all improved, while accuracy has increased from 0.92 in the original dataset to 0.97. Not only having the metrics been improved, but feature selection has also resulted in a computation time of 0.01618 seconds faster than the original dataset. The embedded technique used the shortest duration of time, 0.00619 seconds.

Feature selection (FS) method	Precision	Recall	F1-score	Accuracy	Time(sec)
Without FS	0.94	0.92	0.92	0.92	0.01618
Filter method	0.97	0.97	0.97	0.97	0.01006
Wrapper method	0.97	0.97	0.97	0.97	0.01074
embedded method	0.97	0.97	0.97	0.97	0.00619

Table 7.5: Performance of K-nearest Neighbours classifier according to feature selection

Conclusions

In this section, we studied the impact of feature selection on model performance and computing efficiency and gave the study results in this chapter. We evaluated multiple classifiers. The outcomes revealed the differences between features remaining, accuracy improvements, and processing time. This study's most superb accuracy winner is the Logistic Regression model using the embedded Method, with the highest accuracy score of 0.98. Additionally, it demonstrates quick processing due to the shortest time of 0.00218 seconds.

8.1 Feature selection

Our investigation into three feature selection methods - Filter, Wrapper, and embedded. We will provide a summary as follows.

8.1.1 Filter method

The Filter Method, the Correlation Coefficient-based method, stands out for its computational efficiency, with 0.0035 seconds of processing time and retaining only 8 features. However, it only enhances model performance in the K-Nearest Neighbors (KNN) model. Additionally, it results in improved processing speed for all models except Logistic Regression.

8.1.2 Wrapper method

The Wrapper Method, which uses forward and backward selection, effectively enhances model performance with 14 remaining features. However, it requires the most computational time, totaling 643.82401 seconds, notably more than the embedded Method by around 8 times and significantly greater than the Filter Method by approximately 2×10^6 times. This feature selection technique, in particular, improves the performance of all models except the Naive Bayes classifier.

8.1.3 embedded method

After evaluating the findings, it was discovered that the embedded Method effectively decreased the feature set from 24 to 11 features in 78.92151 seconds, a remarkably short amount of time. Even while it may not result in the most significant feature reduction, it greatly enhances model performance in a shorter time. Using the embedded Method produced the quickest processing time across all models except the Naive Bayes classifier.

Appendix

The use of Python 3 for the study is described in this appendix. Jupyter notebooks were used for the analysis and experimentation, providing an interactive and documented approach to data exploration, model construction, and evaluation.

Visit the following GitHub repository to view the entire codebase, including all scripts and notebooks used in this study: <https://github.com/Sirinja/Dissertation.git>. Thanks to this repository, the methodologies and procedures described throughout the dissertation are transparent and reproducible.

The study's approaches are made accessible and understandable for other academics and practitioners by using Python 3 and Jupyter Notebooks, which align with modern data analysis and machine learning practices.

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